

# Sigrok BMS Protocol Decoder Capture Analysis

Generated from offline sigrok/PulseView analysis outputs

## Executive Summary

**30**

total captures

**17**

RS485/UART captures

**13**

CAN captures

**19 / 6 / 5**

Bridge / Forward / Direct

### Key observations

- Anenji Pylon RS485 JKBMS shows the clearest RS485 latency change: Direct cable is -4.58 ms (-47.2%) versus Bridge for average request-to-response latency.
- Growatt RS485 JKBMS has a much longer Direct cable tail latency: Direct cable is +21.48 ms (+40.1%) versus Bridge for full-exchange P95 latency.
- Growatt CAN SeplosBMS cycles are shorter in Direct cable mode: Direct cable is -164.25 ms (-19.3%) versus Bridge for average CAN cycle duration.
- Growatt CAN JKBMS frame rate stays effectively flat across modes: Direct cable is +0.1 frames/s (+0.2%) versus Bridge.

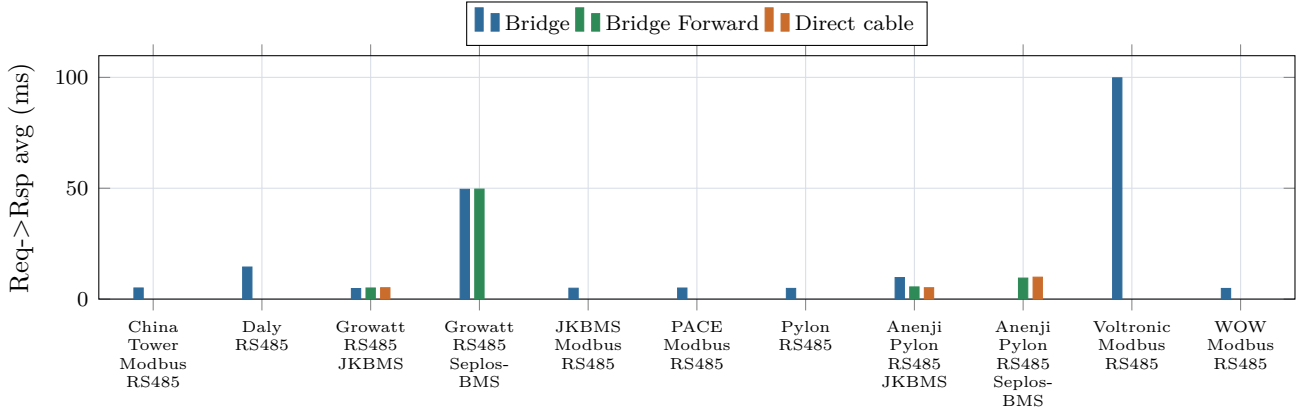
Table 1: Largest Direct cable versus Bridge changes by absolute percentage.

| Group                 | Metric         | Direct vs Bridge  | Abs. change |
|-----------------------|----------------|-------------------|-------------|
| Growatt CAN JKBMS     | Cycles/s       | +0.1 (+100%)      | 100%        |
| Growatt CAN JKBMS     | Cycle min (ms) | -5000.47 (-50.1%) | 50.1%       |
| Growatt CAN JKBMS     | Cycle avg (ms) | -5000.47 (-50.1%) | 50.1%       |
| Growatt CAN JKBMS     | Cycle max (ms) | -5000.47 (-50.1%) | 50.1%       |
| Growatt CAN JKBMS     | Cycle P95 (ms) | -5000.47 (-50.1%) | 50.1%       |
| Growatt CAN SeplosBMS | Cycle min (ms) | -121.3 (-48.3%)   | 48.3%       |

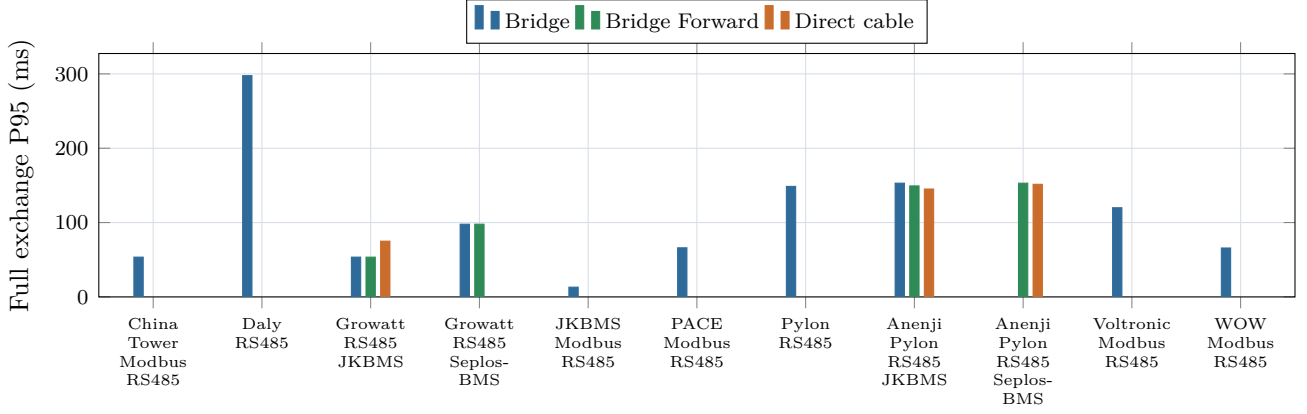
## 1 Protocol Overview Charts

These charts include every protocol group available in the overview CSV. Missing topology modes are intentionally left blank, so a group can have one, two, or three bars depending on the captures currently available.

CAN cycle duration charts use a logarithmic y-axis because values span multiple orders of magnitude, from sub-millisecond single-frame cycles to multi-second capture-length cycles.

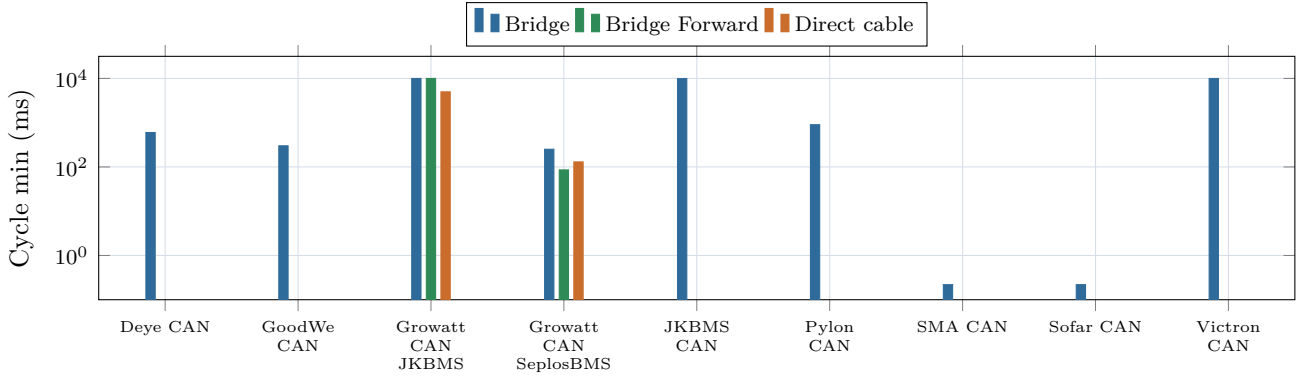


(a) Average request-to-response latency for all available RS485/UART protocol groups.

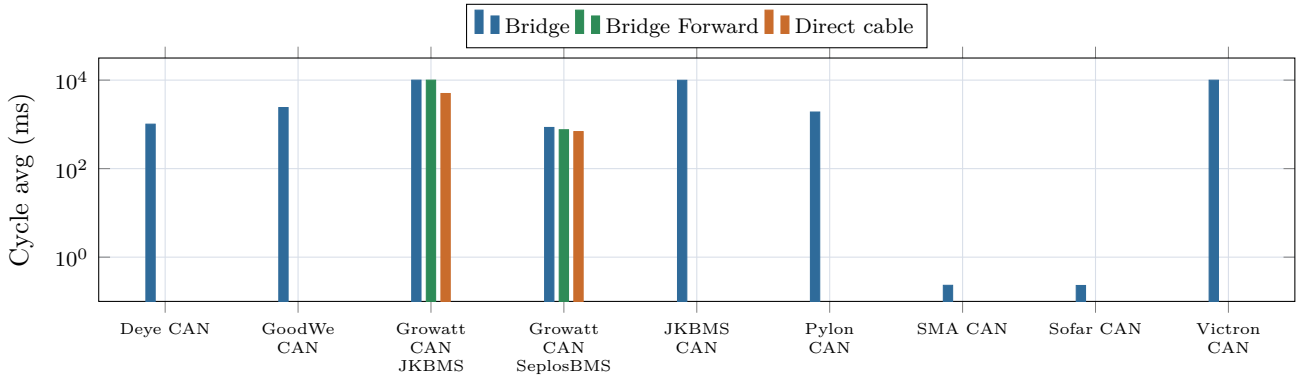


(b) P95 full-exchange duration for all available RS485/UART protocol groups.

Figure 1: RS485/UART protocol overview charts.

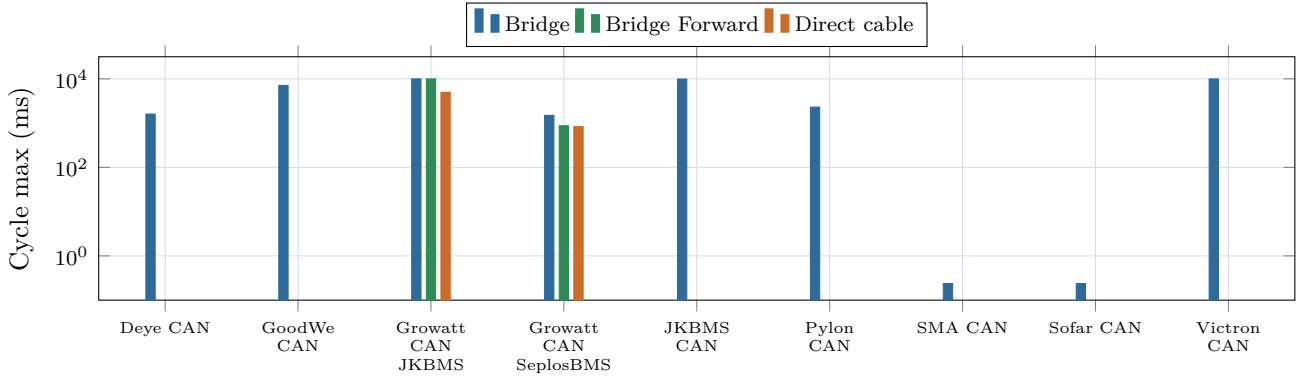


(a) Minimum CAN cycle duration, log scale, for all available CAN protocol groups.

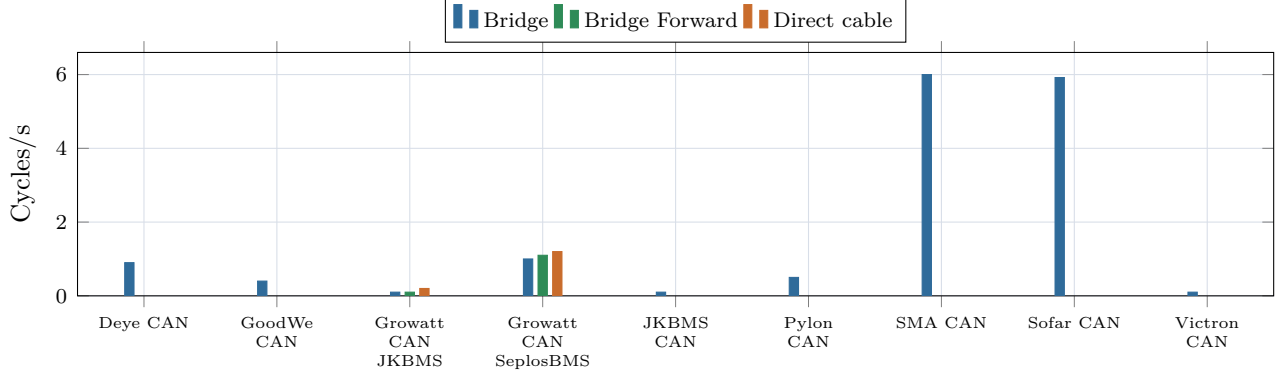


(b) Average CAN cycle duration, log scale, for all available CAN protocol groups.

Figure 2: CAN cycle duration overview charts.

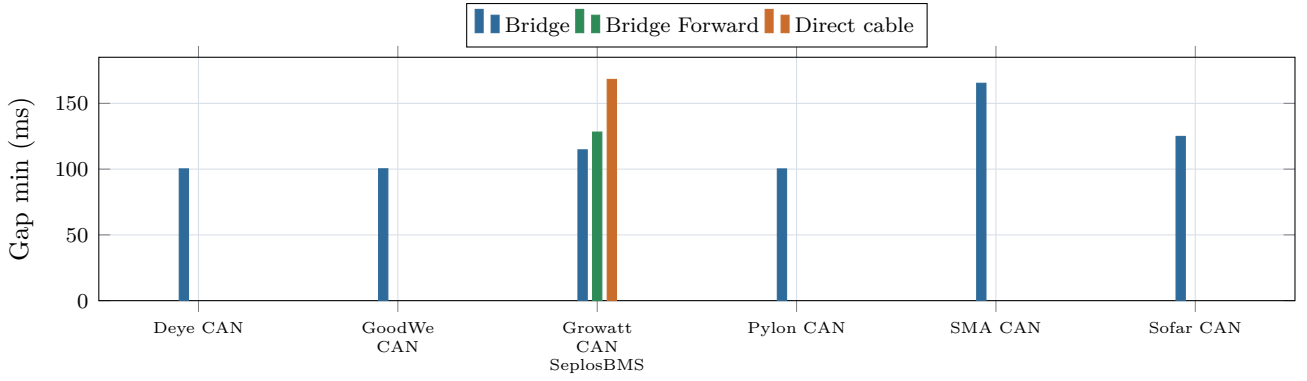


(a) Maximum CAN cycle duration, log scale, for all available CAN protocol groups.

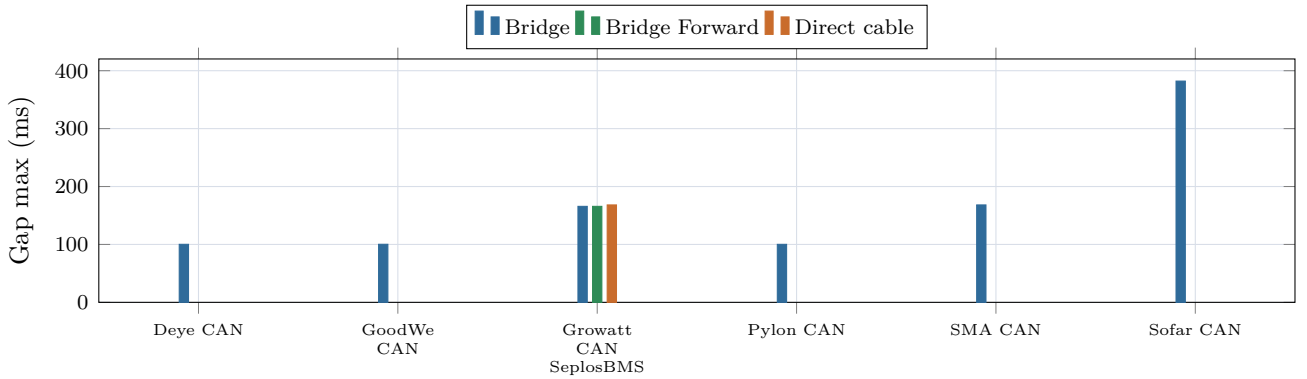


(b) CAN cycle rate normalized by capture duration for all available CAN protocol groups.

Figure 3: CAN maximum cycle duration and throughput charts.



(a) Minimum inter-cycle gap for CAN protocol groups with at least two detected cycles.



(b) Maximum inter-cycle gap for CAN protocol groups with at least two detected cycles.

Figure 4: CAN inter-cycle gap overview charts.

## 2 Three-mode Delta Tables

The following tables focus only on groups where all three modes are available: Bridge, Bridge Forward, and Direct cable. Deltas are directional: orange means an increase, green means a decrease. Whether an increase is good depends on the metric.

### 2.1 RS485/UART Three-mode Deltas

| Group                    | Metric                 | Bridge    | Forward   | Direct    | Forward vs Bridge | Direct vs Bridge   | Direct vs Forward  |
|--------------------------|------------------------|-----------|-----------|-----------|-------------------|--------------------|--------------------|
| Growatt RS485 JKBMS      | Complete exchanges/s   | 2.5       | 2.5       | 2         | 0 (0%)            | -0.5 (-20%)        | -0.5 (-20%)        |
| Growatt RS485 JKBMS      | Req->Rsp avg (us)      | 4760      | 5008.1    | 5126.76   | +248.1 (+5.2%)    | +366.76 (+7.7%)    | +118.66 (+2.4%)    |
| Growatt RS485 JKBMS      | Req->Rsp P95 (us)      | 5294.32   | 5293.78   | 5438.65   | -0.54 (0%)        | +144.33 (+2.7%)    | +144.87 (+2.7%)    |
| Growatt RS485 JKBMS      | Full exchange avg (us) | 49094.79  | 48926.21  | 59337.41  | -168.58 (-0.3%)   | +10242.62 (+20.9%) | +10411.2 (+21.3%)  |
| Growatt RS485 JKBMS      | Full exchange P95 (us) | 53520.07  | 53484.77  | 75000.91  | -35.3 (-0.1%)     | +21480.84 (+40.1%) | +21516.13 (+40.2%) |
| Anenji Pylon RS485 JKBMS | Complete exchanges/s   | 0.8       | 0.8       | 0.8       | 0 (0%)            | 0 (0%)             | 0 (0%)             |
| Anenji Pylon RS485 JKBMS | Req->Rsp avg (us)      | 9706.67   | 5488.62   | 5124.94   | -4218.05 (-43.5%) | -4581.73 (-47.2%)  | -363.68 (-6.6%)    |
| Anenji Pylon RS485 JKBMS | Req->Rsp P95 (us)      | 9912.79   | 6190.08   | 5692.96   | -3722.71 (-37.6%) | -4219.84 (-42.6%)  | -497.12 (-8%)      |
| Anenji Pylon RS485 JKBMS | Full exchange avg (us) | 111164.32 | 106950.36 | 102917.89 | -4213.95 (-3.8%)  | -8246.42 (-7.4%)   | -4032.47 (-3.8%)   |
| Anenji Pylon RS485 JKBMS | Full exchange P95 (us) | 153034.21 | 149320.43 | 145154.53 | -3713.79 (-2.4%)  | -7879.68 (-5.1%)   | -4165.9 (-2.8%)    |

## 2.2 CAN Three-mode Deltas

| Group                  | Metric                   | Bridge     | Forward   | Direct     | Forward vs Bridge   | Direct vs Bridge      | Direct vs Forward      |
|------------------------|--------------------------|------------|-----------|------------|---------------------|-----------------------|------------------------|
| Growatt CAN JKBMS      | Frames/s                 | 51.9       | 52        | 52         | +0.1 (+0.2%)        | +0.1 (+0.2%)          | 0 (0%)                 |
| Growatt CAN JKBMS      | Cycles/s                 | 0.1        | 0.1       | 0.2        | 0 (0%)              | +0.1 (+100%)          | +0.1 (+100%)           |
| Growatt CAN JKBMS      | Cycle min (us)           | 9980699.27 | 9980680.5 | 4980232.08 | -18.77 (0%)         | -5000467.18<br>50.1%) | (- 5000448.42 (-50.1%) |
| Growatt CAN JKBMS      | Cycle avg (us)           | 9980699.27 | 9980680.5 | 4980232.08 | -18.77 (0%)         | -5000467.18<br>50.1%) | (- 5000448.42 (-50.1%) |
| Growatt CAN JKBMS      | Cycle max (us)           | 9980699.27 | 9980680.5 | 4980232.08 | -18.77 (0%)         | -5000467.18<br>50.1%) | (- 5000448.42 (-50.1%) |
| Growatt CAN JKBMS      | Cycle P95 (us)           | 9980699.27 | 9980680.5 | 4980232.08 | -18.77 (0%)         | -5000467.18<br>50.1%) | (- 5000448.42 (-50.1%) |
| Growatt CAN Seplos-BMS | Frames/s                 | 13         | 12.9      | 13         | -0.1 (-0.8%)        | 0 (0%)                | +0.1 (+0.8%)           |
| Growatt CAN Seplos-BMS | Cycles/s                 | 1          | 1.1       | 1.2        | +0.1 (+10%)         | +0.2 (+20%)           | +0.1 (+9.1%)           |
| Growatt CAN Seplos-BMS | Cycle min (us)           | 251234.86  | 85691.12  | 129932.96  | -165543.74 (-65.9%) | -121301.9 (-48.3%)    | +44241.84 (+51.6%)     |
| Growatt CAN Seplos-BMS | Cycle avg (us)           | 851373.72  | 755942    | 687125.94  | -95431.72 (-11.2%)  | -164247.78 (-19.3%)   | -68816.06 (-9.1%)      |
| Growatt CAN Seplos-BMS | Cycle max (us)           | 1496223.99 | 871814.91 | 831769.22  | -624409.08 (-41.7%) | -664454.76 (-44.4%)   | -40045.68 (-4.6%)      |
| Growatt CAN Seplos-BMS | Cycle P95 (us)           | 1221318.48 | 861952.16 | 831768.79  | -359366.32 (-29.4%) | -389549.69 (-31.9%)   | -30183.37 (-3.5%)      |
| Growatt CAN Seplos-BMS | Inter-cycle gap min (us) | 114668.76  | 128174.72 | 168225.68  | +13505.97 (+11.8%)  | +53556.93 (+46.7%)    | +40050.96 (+31.2%)     |
| Growatt CAN Seplos-BMS | Inter-cycle gap avg (us) | 155493.47  | 160226.81 | 168228.48  | +4733.34 (+3%)      | +12735.01 (+8.2%)     | +8001.67 (+5%)         |
| Growatt CAN Seplos-BMS | Inter-cycle gap max (us) | 165777.83  | 165777.83 | 168229.68  | 0 (0%)              | +2451.84 (+1.5%)      | +2451.84 (+1.5%)       |
| Growatt CAN Seplos-BMS | Inter-cycle gap P95 (us) | 165776.23  | 165777.83 | 168229.68  | +1.6 (0%)           | +2453.45 (+1.5%)      | +2451.84 (+1.5%)       |

### 3 Capture Overview

#### 3.1 RS485/UART Capture Overview

| Target                            | Topology        | Frames | Complete | Incomplete | Req->Rsp avg (us) | Req->Rsp P95 (us) | Full P95 (us) |
|-----------------------------------|-----------------|--------|----------|------------|-------------------|-------------------|---------------|
| china tower modbus                | Bridge          | 79     | 39       | 1          | 5015.03           | 5337.93           | 53525.72      |
| daly rs485                        | Bridge          | 88     | 44       | 0          | 14458.98          | 18060.3           | 297646.8      |
| growatt rs485                     | Bridge          | 50     | 25       | 0          | 4760              | 5294.32           | 53520.07      |
| growatt seplos rs485              | Bridge          | 49     | 24       | 1          | 49517.09          | 49863.54          | 97806.11      |
| direct growatt rs485              | Direct cable    | 100    | 50       | 0          | 5126.76           | 5438.65           | 75000.91      |
| forward growatt rs485             | Bridge For-ward | 50     | 25       | 0          | 5008.1            | 5293.78           | 53484.77      |
| forward growatt seplos rs485      | Bridge For-ward | 49     | 24       | 1          | 49604.18          | 50490.32          | 97846.4       |
| jkbms modbus                      | Bridge          | 78     | 39       | 0          | 4903.5            | 5191.55           | 13115.59      |
| pace modbus                       | Bridge          | 79     | 39       | 1          | 4987.74           | 5260.88           | 66064.78      |
| pylon rs485                       | Bridge          | 49     | 18       | 5          | 4817.5            | 5537.46           | 148667.74     |
| anenji pylon rs485                | Bridge          | 20     | 8        | 4          | 9706.67           | 9912.79           | 153034.21     |
| direct anenji jkbms pylon rs485   | Direct cable    | 61     | 20       | 21         | 5124.94           | 5692.96           | 145154.53     |
| direct anenji seplos pylon rs485  | Direct cable    | 50     | 20       | 10         | 9887.21           | 12043.43          | 151498.7      |
| forward anenji pylon rs485        | Bridge For-ward | 24     | 8        | 8          | 5488.62           | 6190.08           | 149320.43     |
| forward anenji seplos pylon rs485 | Bridge For-ward | 20     | 8        | 4          | 9469.73           | 9877.58           | 153001.64     |
| voltronic modbus                  | Bridge          | 78     | 39       | 0          | 99796.75          | 101837.19         | 119962.1      |
| wow modbus                        | Bridge          | 78     | 39       | 0          | 4799.13           | 5093.6            | 65808.75      |

## 3.2 CAN Capture Overview

| Target                    | Topology        | Frames | IDs | Cycles | Cycle min (us) | Cycle avg (us) | Cycle max (us) | Cycle P95 (us) | Errors |
|---------------------------|-----------------|--------|-----|--------|----------------|----------------|----------------|----------------|--------|
| deye can                  | Bridge          | 100    | 8   | 9      | 599848.77      | 1011013.33     | 1599872.42     | 1599871.23     | 0      |
| goodwe can                | Bridge          | 100    | 8   | 4      | 300245.9       | 2399973.05     | 7100062.15     | 6275024.24     | 0      |
| growatt can               | Bridge          | 519    | 10  | 1      | 9980699.27     | 9980699.27     | 9980699.27     | 9980699.27     | 0      |
| growatt sepos can         | Bridge          | 130    | 12  | 10     | 251234.86      | 851373.72      | 1496223.99     | 1221318.48     | 0      |
| direct growatt can        | Direct cable    | 260    | 10  | 1      | 4980232.08     | 4980232.08     | 4980232.08     | 4980232.08     | 0      |
| direct growatt sepos can  | Direct cable    | 65     | 12  | 6      | 129932.96      | 687125.94      | 831769.22      | 831768.79      | 0      |
| forward growatt can       | Bridge For-ward | 520    | 10  | 1      | 9980680.5      | 9980680.5      | 9980680.5      | 9980680.5      | 0      |
| forward growatt sepos can | Bridge For-ward | 129    | 12  | 11     | 85691.12       | 755942         | 871814.91      | 861952.16      | 0      |
| jkbms can                 | Bridge          | 344    | 12  | 1      | 9907992.71     | 9907992.71     | 9907992.71     | 9907992.71     | 0      |
| pylon can                 | Bridge          | 100    | 8   | 5      | 900275.94      | 1900011.26     | 2299947.27     | 2299945.2      | 0      |
| sma can                   | Bridge          | 60     | 6   | 60     | 219.4          | 230.73         | 237.4          | 237.4          | 0      |
| sofar can                 | Bridge          | 148    | 6   | 148    | 219.4          | 228.76         | 237.4          | 237.4          | 0      |
| victron can               | Bridge          | 250    | 19  | 1      | 9960725.84     | 9960725.84     | 9960725.84     | 9960725.84     | 0      |

## Notes

- The LaTeX source is generated from committed CSV files under `analysis/out/`.
- Capture durations are not identical, so normalized rates are included where useful.
- CAN cycle duration can be dominated by capture length when only one cycle is detected.
- Compiled PDFs and LaTeX build artifacts should stay out of the repository.